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Design and Implementation of an ArduSub-Based Unmanned Surface Vehicle with Telemetry and Accelerometer Calibration

Wennan Zhu [1], Yu Liu [1,2,*] and Weicheng Cui [1,2,*]

[1] School of Engineering, Westlake University, Hangzhou 310030, China; zhuwennan@westlake.edu.cn
[2] Key Laboratory of 3D Micro/Nano Fabrication and Characterization of Zhejiang Province, Hangzhou 310030, China
* Correspondence: liuyu36@westlake.edu.cn (Y.L.); cuiweicheng@westlake.edu.cn (W.C.)

Abstract

The frontiers of oceanographic survey and environmental monitoring have been expanded with the rapid development of Unmanned Surface Vehicles (USVs). This report presents the design and construction of a USV based on the ArduSub firmware. Through a stable control architecture and MAVLink communication, the system can be customized for surface applications. The proposed system consists of a modular vehicle design, a high-rate data transmission link implemented using the Pymavlink, and a remote control via the standard joystick protocols. In order to enhance the performance of motion estimation, the triaxial accelerometer is calibrated using the nine-parameter ellipsoid fitting technique, which removes bias, scale factors, and axis misalignment errors. The system is also scalable, providing opportunities for future development in autonomous navigation and multi-sensor integration.

Keywords: Unmanned Surface Vehicle (USV); ArduSub; system integration; accelerometer calibration

1. Introduction

Unmanned Surface Vehicles (USVs) have become increasingly important in recent years due to the increased need for scalable, safe, and long-term ocean monitoring. Traditional monitoring approaches are often constrained by high operational costs and risks to human operators, especially under extreme sea conditions. USVs provide a viable substitute, as they can be used to operate continuously in harsh or remote locations, and yet, real-time data can be obtained to perform tasks like environmental monitoring, bathymetric surveys and marine ecological research. Moreover, their relatively simple design and maintenance make them suitable for rapid prototyping and experimental validation in research contexts.

Open-source autopilot frameworks have also contributed to the rapid development of research USVs, and have greatly simplified system integration and design. ArduSub is a submodule of the ArduPilot ecosystem that is specifically designed for underwater vehicles [1]. It offers a stable and well-tested control architecture, flexible motor mixing and native MAVLink communication. These features make ArduSub compatible with USVs, especially those that require a multi-thruster layout to enable better maneuvering and heading control. Moreover, it can be easily integrated with standard ground control stations and other companion computers (such as Raspberry Pi), allowing it to be used in both teleoperated experiments and future autonomous operations [2].

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Despite these advantages, challenges remain in ensuring reliable onboard sensing, particularly for inertial measurements, which are critical for motion estimation and control performance. Sensor imperfections, such as bias and scaling errors, can significantly degrade system accuracy if not properly addressed.

This report presents the practical development of an ArduSub-based USV from a system integration perspective. Rather than targeting a fully autonomous solution, this work focuses on constructing a flexible experimental platform that integrates mechanical modularity, real-time communication, and data transmission. The main contribution of this study lies in the implementation of a low-cost and compact ArduSub-based USV and the evaluation of an accelerometer calibration method based on an ellipsoid fitting method, aimed at improving the reliability of onboard inertial sensing.

The remainder of the paper is structured as follows. Section 2 reviews the background of USVs and ArduSub. Section 3 describes the design of the proposed USV. Section 4 presents the mathematical model used for calibration and acceleration estimation. Experimental analysis and results are provided in Section 5, followed by a discussion of system performance in Section 6. Section 7 concludes the paper.

2. Development of USV and ArduSub

Advances in open-source autopilot software have been closely related to the development of marine robotics and unmanned vehicles. Initially, ArduPilot was a community project intended to serve low-cost model aircraft and rovers on the ground. Over time, with advances in hardware architecture and modular software design, these systems were expanded to also support other types of vehicles, such as boats and submersibles [3]. Open-source autopilot ecosystems have become useful research platforms due to their capability to support companion computers, middleware (including ROS, the Robot Operating System), and simulation platforms (including software and hardware-in-the-loop testing). These qualities allow for quick prototyping and testing of autonomous systems [4].

ArduRover is a widely used branch of the ArduPilot ecosystem applied to surface vehicles, and supports wheeled and marine vehicles with simple navigation capabilities [5]. Nevertheless, its kinematic models (e.g., differential steering and non-holonomic motion) do not apply as well to craft with vectored propulsion, which require coordinated lateral force generation and heading control. Thus, ArduSub has been considered another control framework by researchers and developers. ArduSub was originally developed for underwater operation but has proven to be usable in other applications due to its openness to multiple thruster designs and its modular control design [6,7]. Recent studies also added real hydrodynamic parameters to ArduSub simulations. This improves the accuracy of controller testing for marine vehicles [8].

Concurrently, the state estimation of USVs has been receiving growing research interest, particularly when subjected to environmental perturbations as well as sensing constraints. For example, nonlinear extended state observers have been designed to reconstruct unmeasured propulsion, lateral, and yaw velocities and at the same time estimate unknown disturbances that could be caused by wind, waves, and hydrodynamic uncertainties. Its stability has been verified using Lyapunov analysis [9]. Other approaches include GPS/IMU integration, which can provide robust estimation of position and velocity despite the presence of measurement noise [10]. This highlights the effectiveness of the loosely coupled sensor integration strategies.

Other, more recent research has been focused on robustness and resilience in complex operating environments. Challenges in motion state estimation and resilience to network attacks on wireless sensor networks were solved through a 2024 set-membership filtering framework [11]. It showed that there is a way to ensure reliable performance despite

being impaired under communication degradation. On the same note, recent research has suggested the use of zonotopic reachability-based estimation techniques that can maintain accurate state estimates even in the presence of sensor channels that are manipulated maliciously [12].

There has also been increasing development in multi-sensors and multi-robots. One of the studies, which was conducted in 2025, suggested a nonlinear six-degrees-of-freedom model with sensor fusion between the UAV and USV platforms. It achieves higher accuracy of estimation and sound prediction of the motion of vessels in turbulent sea conditions, which is tested in simulations and in real-world experiments [13]. Similar studies have already shown that in the absence of or interference with a Global Navigation Satellite System (GNSS), the visual measurement data of unmanned aerial vehicles can be incorporated into an extended Kalman filter framework to position USVs [14]. It offers an encouraging alternative method of navigation under limited conditions.

In the meantime, the general use of Micro-Electro-Mechanical System (MEMS) sensors has contributed significantly to lowering the price and volume of navigation and inertial sensing systems. MEMS accelerometers and gyroscopes are widely used in robotics and unmanned systems because of their small footprint, low power usage, and capability to make measurements with high frequency, making real-time motion estimation on platforms such as drones and USVs possible.

3. Design

3.1. Hardware Architecture Design

The hardware architecture of the proposed USV is designed with a compact and functionally layered architecture, which provides reliable control, effective communication, and stable power delivery. The propulsion subsystem comprises two electric thrusters that are used to generate lateral force for directional control and maneuvering. These are driven and controlled by the Pixhawk autopilot. It enables precise thrust control and responds to commands in real time.

The Pixhawk 2.4.8 autopilot is the main control, processing, and stabilization unit in this paper. It acquires inertial, positional, and navigation data; executes control algorithms; and generates actuator signals for the propulsion system. An external GPS receiver provides absolute positioning and velocity data, which are used to aid in navigation and state estimation when operating.

The onboard companion computer is a Raspberry Pi 3B. It performs high-level tasks like data processing, telemetry processing, and communication management. It is connected to the Pixhawk 2.4.8 autopilot through a USB connection to communicate the navigation data and the control information. A single battery pack supplies power, and power is regulated to provide a stable voltage to low-power logic modules and high-current propulsion devices.

This integrated and scalable hardware design ensures the convenient maintenance of the system and facilitates component replacement, enabling the incorporation of additional sensors or independent movement capabilities. Such attributes make the platform suitable for experimental validation and research. The hardware architecture is shown in Figure 1.

3.2. Software Design and Communication Protocols

The proposed USV system adopts a computational architecture that integrates real-time control, sensor data acquisition, and telemetry transmission in a thread-safe and modular way. The vehicle is operated using a Pixhawk autopilot, which is connected to the DroneKit-Python (version 2.9.1) API through a serial connection (/dev/ttyUSB0) and a baud rate of 115,200. DroneKit is a Pymavlink-based module [15]. The MAVLink-based

communication system provides a reliable communication with an average end-to-end latency of 10–150 ms, depending on network conditions. It achieves a reliable range of approximately 200 m in open-water environments. This configuration enables the flight mode of the vehicle to be controlled, as well as the arming status, channel overrides, and access to data from onboard sensors (velocity, attitude, raw IMU measurements, etc.) on a high level. Figure 2 illustrates the main structure of communication.

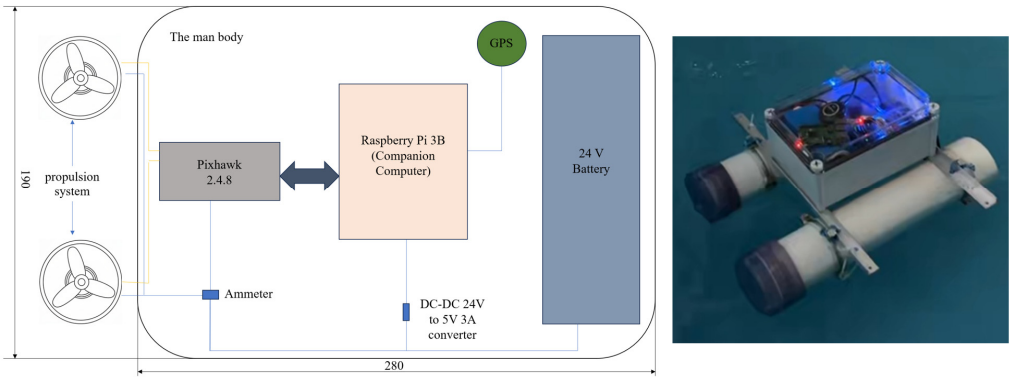


Figure 1. The Description of the Hardware Architecture.

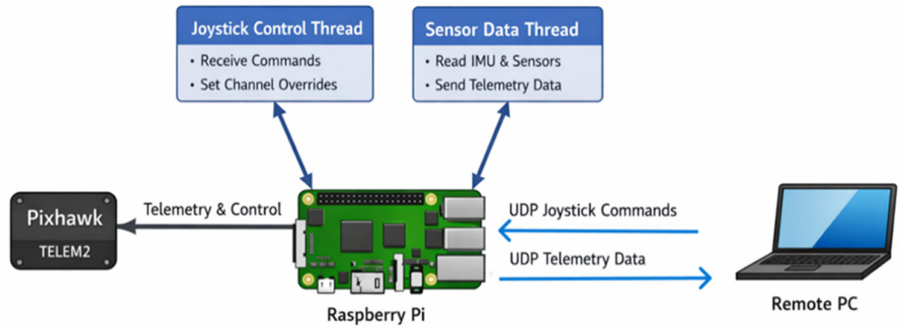


Figure 2. The Principle of Data and Command Transmission.

In order to realize responsive remote operation, the system uses a two-threaded architecture. Joystick commands are handled by a dedicated thread (joystick thread) continuously on a specific port via UDP, and the channel values are converted into corresponding PWM override signals for the Pixhawk. This approach provides manual inputs to the control in almost real time without interfering with other processes.

At the same time, a second thread (sensor thread) is used to obtain and transmit the sensor data. It receives the Pixhawk messages of RAW IMU to access the accelerometer readings, which are converted into physical units (m/s^2). The thread, together with IMU data, collects the current flight mode and its arming status; the attitude roll, pitch, and yaw values; and the depth and velocity. All the sensor values are compiled into a message in JSON format and transmitted via UDP to a host PC at a fixed interval of 0.25 s (4 Hz) via a second UDP socket.

The use of non-blocking UDP for both command reception and telemetry transmission helps to minimize latency and prevent conflicts between control and data streams. Separating control and sensing threads improves system reliability since joystick inputs are handled differently from data collection and data transmission. In general, this design is well designed, flexible, and easily extended to accommodate other sensors or autonomous modules in the future, in addition to being capable of supporting real-time remote control and high-rate data acquisition.

3.3. Main Specifications

To improve clarity and reproducibility, the main specifications of the proposed USV platform are summarized in Table 1.

Table 1. Summary of USV Platform Specifications.

Category	Component/Parameter	Specification	Description
Control Unit	Flight Controller	Pixhawk 2.4.8	Main control unit for navigation, stabilization, and actuator control
Companion Computer	Onboard Processor	Raspberry Pi 3B	Handles high-level computation, telemetry, and communication
Navigation Sensors	GPS Module	UBLOX NEO-M8N GPS Module	Provides global position and velocity for navigation
Inertial Sensors	IMU	Built-in Pixhawk IMU	Measures acceleration, angular velocity, and attitude
Propulsion System	Thrusters	ROVMAKER 2216 brushless thrusters ($\times$2)	Provides differential thrust for steering and maneuvering
Power Supply	Battery	24 V, 10 Ah lithium battery pack	Provides power for propulsion and onboard electronics
Power Regulation	Voltage Regulation	DC-DC converters	Ensures stable voltage for logic and high-current devices
Communication 1	Protocol	MAVLink (via DroneKit-Python)	Enables control commands and telemetry exchange
Communication 2	Interface	USB	Serial communication between Pixhawk and Raspberry Pi
Telemetry Transmission	Data Link	UDP (JSON format, 4 Hz)	Sends sensor and status data to ground station

4. Mathematical Modeling of Acceleration Calibration and Estimation

4.1. Coordinate Reference Systems

Marine vehicles are typically analyzed using two orthogonal coordinate systems to describe vehicle motion relative to Earth and the vehicle frame.

The former is the Earth-fixed North–East–Down (NED) frame, which is considered an inertial frame in small distances under the flat-Earth approximation. The x-axis points toward the geographic north, the y-axis is the geographic east, and the z-axis is downwards in the direction of gravity. This convention is widely used in marine and aerospace navigation, as it offers a right-handed coordinate system and an intuitive way of describing position and orientation on Earth [16].

The second frame is the body-fixed frame, which moves and rotates with the vehicle. Its origin is typically the vehicle's center of gravity (CG), and its axes are aligned with the principal directions of the vehicle: surge (forward), sway (lateral), and heave (vertical). Velocities expressed in this frame represent instantaneous motion relative to the craft's orientation, rather than global position. Figure 3 illustrates the NED frame compared to the body-fixed frame.

In order to relate the velocity in the body coordinate system to the motion in the inertial NED coordinate system, the vehicle attitude is usually used for rotational transformation.

The attitude is typically represented by yaw, pitch, and roll angles (ψ, θ, ϕ). The rotation matrix from body frame to NED frame is defined using a ZYX Euler sequence:

$$R = R_z(\psi)R_y(\theta)R_x(\phi) \tag{1}$$

$$R_x(\phi) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & cos\,\phi & -sin\,\phi \\ 0 & sin\,\phi & cos\,\phi \end{bmatrix} \tag{2}$$

$$R_y(\theta) = \begin{bmatrix} cos\,\theta & 0 & sin\,\theta \\ 0 & 1 & 0 \\ -sin\,\theta & 0 & cos\,\theta \end{bmatrix} \tag{3}$$

$$R_z(\psi) = \begin{bmatrix} cos\,\psi & -sin\,\psi & 0 \\ sin\,\psi & cos\,\psi & 0 \\ 0 & 0 & 1 \end{bmatrix} \tag{4}$$

and,

$$\begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{z} \end{bmatrix} = R \begin{bmatrix} u \\ v \\ w \end{bmatrix} \tag{5}$$

where $[x, y, z]^\top$ is the position vector in the NED frame and $[u, v, w]^\top$ is the velocity vector in the body frame.

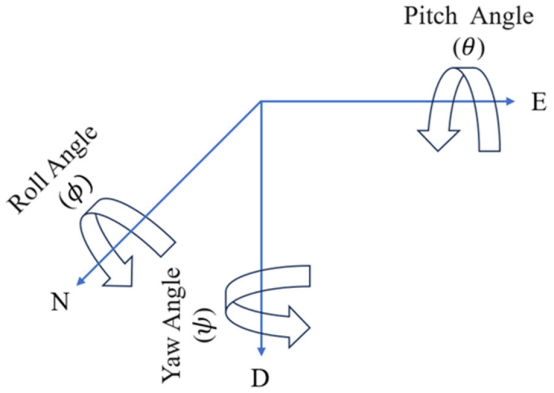


Figure 3. NED Frame with Rotational Angles.

4.2. Sensor Error Models

To obtain accurate acceleration for the kinetic equations, the raw measurements from a triaxial MEMS accelerometer must be corrected for systematic errors. A widely accepted error model for a three-axis accelerometer relates the measured output a_{meas} to the true acceleration a_{true} through sensor imperfections as follows:

$$a_{meas} = SMa_{true} + b + n \tag{6}$$

In this model, the measured acceleration vector a_{meas} is affected by several sources of error. The scale factor matrix S represents sensitivity variations along each axis, while the misalignment or orthogonality matrix M accounts for any deviations of the sensor axes from perfect orthogonality. A static bias vector b models the zero-gravity offset present even when the sensor is stationary, and n captures random noise, usually assumed to be Gaussian. Finally, the gravity vector g must be compensated for or removed when the measured acceleration includes the effect of gravity. These error terms characterize the

difference between actual acceleration and sensor output and are used in the calibration process, including ellipsoid fitting.

When the device is static, it is affected only by gravitational acceleration. The ideal magnitude of the true acceleration vector is equal to the local gravitational acceleration $\| g \|$. In practice, due to bias, scale, and misalignment errors, the measured 3D points scatter around a distorted ellipsoid in $\mathbb{R}^3$ rather than lying on a perfect sphere of radius $\| g \|$. Calibration therefore seeks to determine matrices **S**, **M**, and vector **b** such that the corrected measurements satisfy:

$$||a_{true}|| \approx ||g|| \tag{7}$$

To be more convenient, it can also be written as:

$$||a_{true}||^2 \approx ||g||^2 \tag{8}$$

This geometric constraint is the basis for ellipsoid-fitting calibration algorithms that solve for the calibration parameters without requiring expensive precision turntables or reference platforms. By collecting measurements of the sensor in multiple static orientations and fitting an ellipsoid to the raw data, the systematic errors can be estimated and eliminated [17,18].

4.3. IMU Calibration: The Least-Squares Ellipsoid Method

The accelerometer calibration is based on an ellipsoid fitting approach, which models the sensor measurements as a distorted ellipsoid due to bias, scale factor, and axis misalignment errors. This formulation has been widely adopted in the calibration of triaxial sensors, where the raw measurements are constrained to lie on an ellipsoid surface rather than an ideal sphere. By estimating the ellipsoid parameters and applying matrix decomposition, the intrinsic calibration matrix can be recovered. Similar approaches have been extensively studied in the literature for IMU and magnetometer calibration, demonstrating their effectiveness for low-cost MEMS sensors [19–22].

Low-cost MEMS accelerometers suffer from scale factor errors, axis misalignment, and bias offsets. From Equation (6), the inverse model can be derived as:

$$a_{true} = S_{inv}(a_{meas} - C) \tag{9}$$

Thus,

$$S_{inv} = (SM)^{-1},\ C = b \tag{10}$$

where $\mathbf{a}_{\text{true}}$ is the true acceleration vector, $\mathbf{a}_{\text{meas}}$ is the raw accelerometer measurement, **C** denotes the bias term (including constant bias and measurement noise), and $\mathbf{S}_{\text{inv}}$ is the inverse sensitivity matrix that compensates for scale factor and non-orthogonality errors. Based on Equation (8), we obtain:

$$(a_{meas} - C)^T {S_{inv}}^T S_{inv}(a_{meas} - C) = ||g||^2 \tag{11}$$

$${a_{meas}}^T {S_{inv}}^T S_{inv} a_{meas} - 2C^T {S_{inv}}^T S_{inv} a_{meas} + C^T {S_{inv}}^T S_{inv} C = ||g||^2 \tag{12}$$

When the IMU is placed in static conditions, the only acceleration acting on the sensor is gravity. Therefore, the magnitude of the true acceleration satisfies Equation (8).

Due to sensor imperfections, the measured acceleration points do not lie on a sphere but instead form an ellipsoid in the measurement space. Under static conditions, each accelerometer sample $x = (x, y, z)^T$ approximately satisfies a general ellipsoid equation:

$$a_1x^2 + a_2y^2 + a_3z^2 + a_4xy + a_5xz + a_6yz + a_7x + a_8y + a_9z = 1 \tag{13}$$

Instead of solving a fully nonlinear calibration problem, this representation allows the unknown coefficients to be estimated in a linear form.

Equation (13) can be rewritten as:

$$da = 1 \tag{14}$$

where $d = \begin{bmatrix} x^2 & y^2 & z^2 & xy & xz & yz & x & y & z \end{bmatrix}$, $a = \begin{bmatrix} a_1 & a_2 & a_3 & a_4 & a_5 & a_6 & a_7 & a_8 & a_9 \end{bmatrix}^T$.

With N static samples, stacking all measurements yields:

$$Da = 1 \tag{15}$$

where $D \in \mathbb{R}^{N \times 9}$ and a_{10} is an $N \times a_{10}$ vector of ones.

The residual vector is defined as:

$$r = Da - 1 \tag{16}$$

and the parameter vector a is obtained by minimizing:

$$\min_a |r|_2^2 = \min_a |Da - 1|_2^2 \tag{17}$$

The least-squares solution can be written as:

$$a = \left(D^T D\right)^{-1} D^T 1 \tag{18}$$

Equation (13) can also be written as:

$$x^T Q x - 2I^T Q x = 1 \tag{19}$$

In this equation, Q, I can be represented as:

$$I = -\frac{1}{2}(Q)^{-1} \begin{bmatrix} a_7 & a_8 & a_9 \end{bmatrix}^T \tag{20}$$

$$Q = \begin{bmatrix} a_1 & \frac{a_4}{2} & \frac{a_5}{2} \\ \frac{a_4}{2} & a_2 & \frac{a_6}{2} \\ \frac{a_5}{2} & \frac{a_6}{2} & a_3 \end{bmatrix} \tag{21}$$

To make Equation (19) similar to Equation (12), it can be written as:

$$(x - I)^T Q (x - I) = 1 + I^T Q I = \frac{||g||^2}{k} \tag{22}$$

In order to transform Equation (19) to Equation (12), a scaling element k is added:

$$k = \frac{||g||^2}{1 + I^T Q I} \tag{23}$$

The symmetric matrix Q is subjected to eigenvalue decomposition (EVD) as shown below:

$$Q = UVU^T \tag{24}$$

Here, U is the matrix of eigenvectors. V is a diagonal matrix that contains the eigenvalues.

The inverse sensitivity matrix used in the calibration process is:

$$S_{inv} = \sqrt{k} U V^{\frac{1}{2}} U^T \tag{25}$$

The bias accelerator C is given as:

$$C = -\frac{1}{2}(Q)^{-1} \begin{bmatrix} a_7 & a_8 & a_9 \end{bmatrix}^T \tag{26}$$

The matrix S_{inv} and the bias vector **C** are applied to the raw accelerometer data. This transformation changes the ellipsoid distribution of the measurements into a sphere with radius g. The process corrects bias, scale error, and axis misalignment in the sensor measurements.

5. Experimental Analysis

The IMU calibration and gravity compensation pipeline performance was evaluated through simulation and real-world experiments. The validation includes simulation tests, static calibration experiments, and dynamic motion tests on a USV platform. The evaluation focuses on measurement consistency, alignment with gravity, and linear acceleration reconstruction accuracy.

5.1. Simulation Test

Before experiments in the real world, a simulation to test the calibration and gravity compensation pipeline was conducted. Known vehicle orientations and motions were synthetically measured using accelerometers. Noise similar to that of the Pixhawk IMU was introduced randomly to simulate actual sensor behavior. To make the outcome more evident, the large bias and noise were selected:

$$a = R \cdot g + b + sigma \cdot k \tag{27}$$

in which R is a unit direction vector representing sensor orientation, g is the gravity magnitude, b represents the fixed offset for each axis, sigma is the standard deviation per axis, and k is a random variable.

The simulated data were processed using ellipsoid fitting. To ensure reproducibility, the simulation parameters were explicitly defined. The g was set to 9.80665 m/s^2. The bias vector was set to:

$$b = [0.8, -0.6, 1.2] \tag{28}$$

and the large standard deviation of the Gaussian noise for each axis was:

$$sigma = [2, 6, 5] \tag{29}$$

The random variable k follows a zero-mean unit Gaussian distribution, i.e., $k \sim N(0, 1)$. On calibration, the mean difference between the magnitude of acceleration and the reference gravity was reduced to 2.5414 m/s^2 compared to 3.7003 m/s^2 before calibration. The main reason for the remaining large error may be due to the large sigma. To be closer to reality, a second test was conducted. The accelerometer cloud is presented in Figure 4, and the acceleration norm error is illustrated in Figure 5.

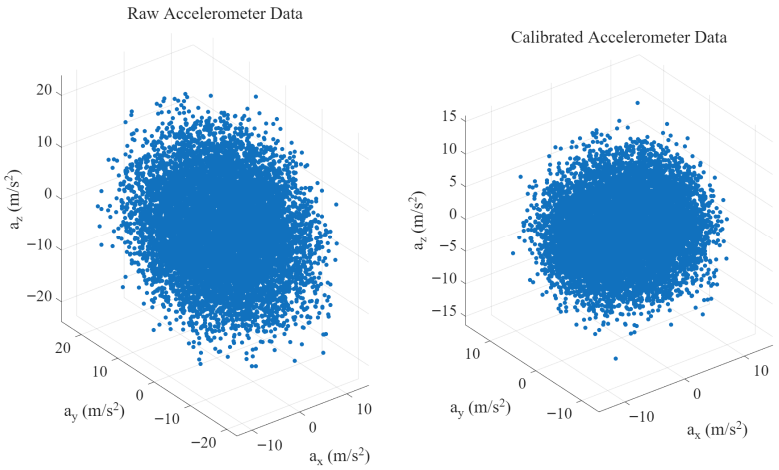


Figure 4. The Raw Accelerometer Cloud and Calibrated Accelerometer Cloud of the Simulation Test of the First Test.

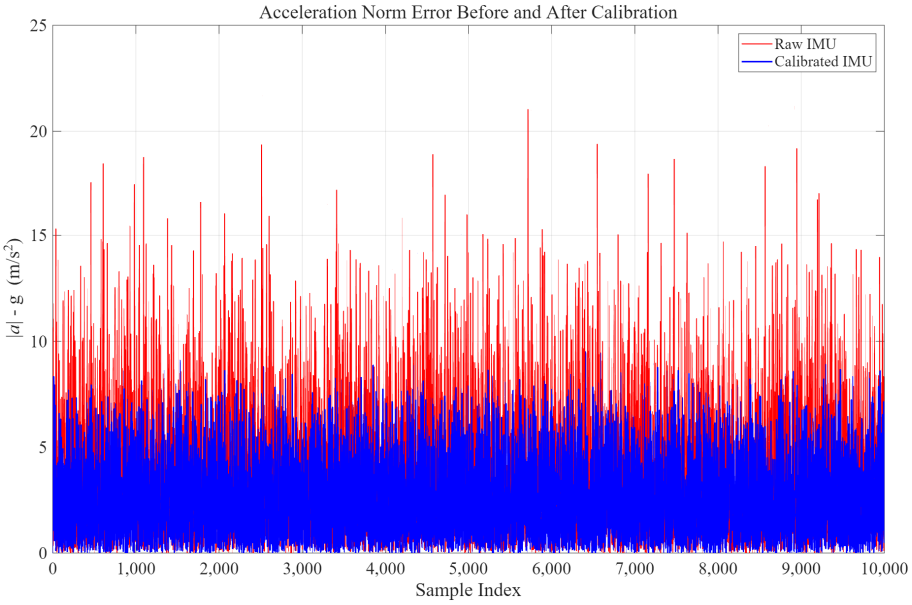


Figure 5. Acceleration Norm Error of the Simulation Test Before and After Calibration of First Test.

In the second test, the bias vector was the same as the first one, and the small standard deviation of the Gaussian noise for each axis was:

$$sigma = [0.2, 0.2, 0.2]. \tag{30}$$

After ellipsoid calibration, the mean absolute error of the acceleration norm decreased from 0.7972 m/s^2 to 0.1610 m/s^2, corresponding to a 79.80% reduction in gravity magnitude estimation error. This indicates that the calibration and gravity compensation pipeline worked as required. The accelerometer cloud is presented in Figure 6, and the acceleration norm error is illustrated in Figure 7.

The resultant discrepancy was explained by sensor noise and minor attitude estimation errors. The data presented by the simulation were used as a baseline to interpret the experimental data in the real world.

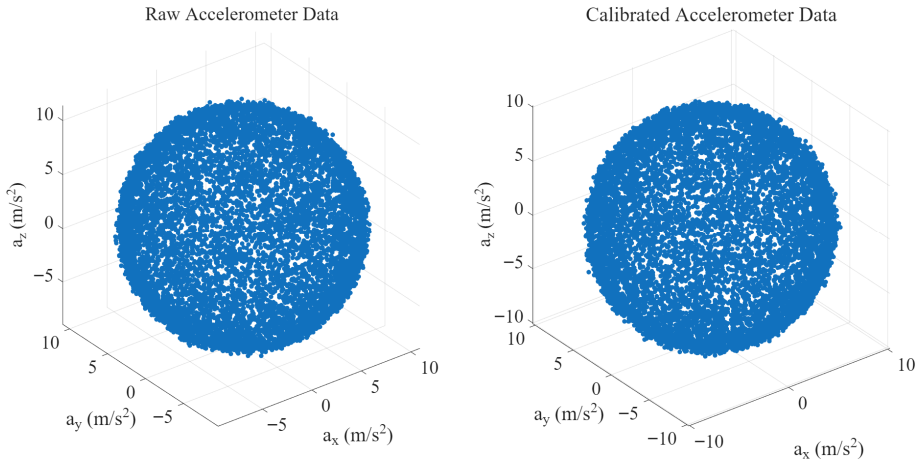


Figure 6. The Raw Accelerometer Cloud and Calibrated Accelerometer Cloud of the Simulation Test of the Second Test.

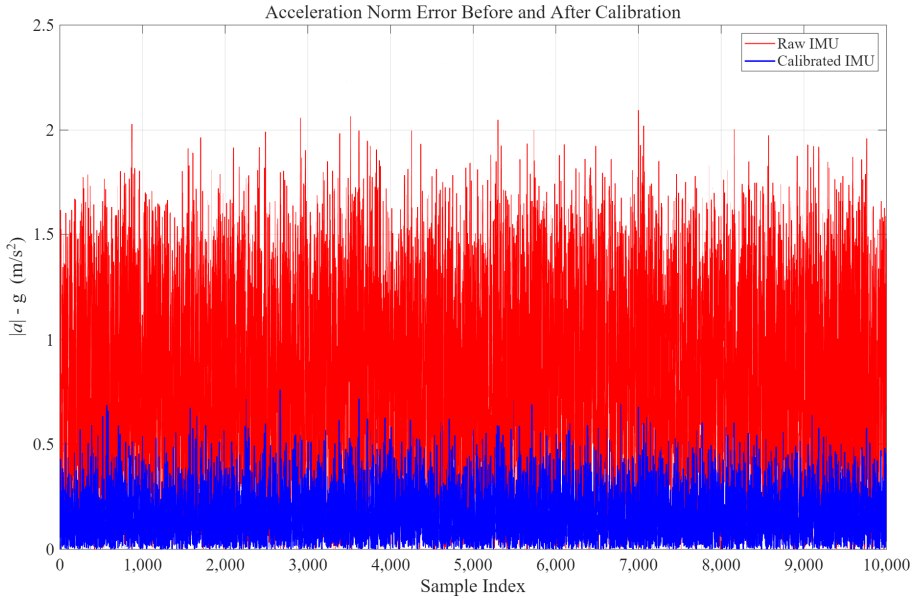


Figure 7. Acceleration Norm Error of the Simulation Test Before and After Calibration of the Second Test.

5.2. Static Calibration

The IMU was placed in various orientations while the USV was kept stationary. Measurements were taken over a few seconds so as to minimize random variations. The local gravitational acceleration was assumed to be $g = 9.80665$ m/s^2. Prior to calibration, the mean absolute error between the measured magnitude of acceleration and gravitational acceleration (g) was about 0.2946 m/s^2. After ellipsoid fitting calibration, this was reduced to 0.2647 m/s^2, corresponding to a 10.2% improvement. The accelerometer cloud is displayed in Figure 8, and the difference in the acceleration error norm is displayed in Figure 9.

To assess whether this improvement is statistically significant rather than arising from stochastic sensor noise, a paired t-test was performed on the sample-wise errors. The results show a highly significant difference between the two conditions ($t = -21.68$, $p < 0.001$). Furthermore, the 95% confidence interval of the mean difference is $[-0.0326, -0.0272]$, which does not include zero, indicating a consistent reduction in error across all samples.

These results demonstrate that the observed improvement is not attributable to random fluctuations in sensor noise, but instead reflects a systematic reduction in measurement error induced by the proposed calibration method.

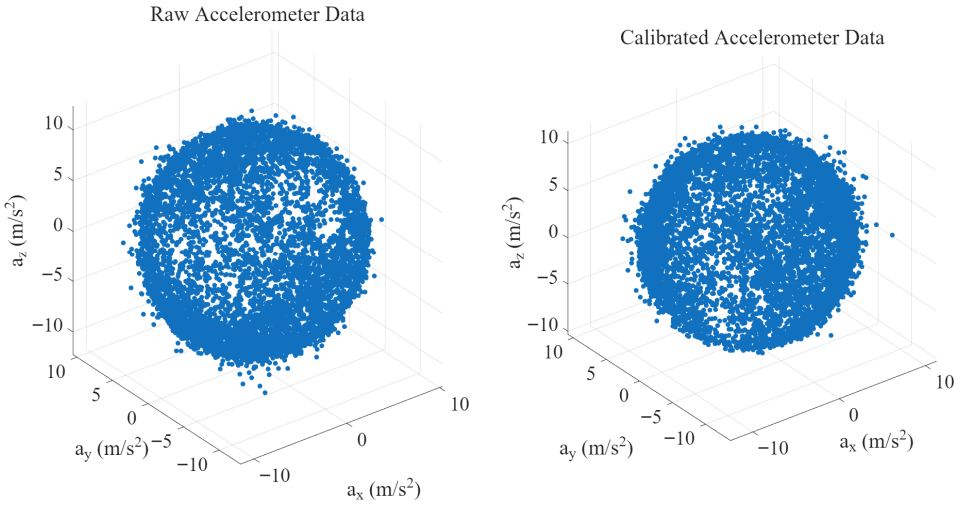


Figure 8. The Raw Accelerometer Cloud and Calibrated Accelerometer Cloud of the Static Test.

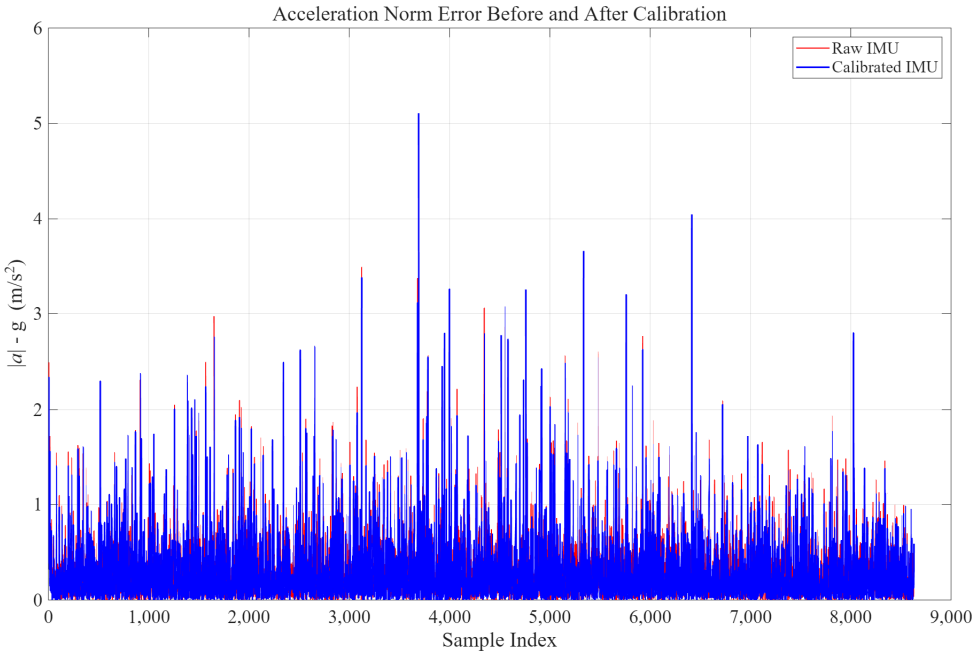


Figure 9. Acceleration Norm Error of the Static Test Before and After Calibration.

This decrease indicates that certain bias and scale factor errors were compensated for. The remaining discrepancies were caused by sensor noise, small misalignment of the IMU and small errors in the attitude estimates provided by the extended Kalman filter (EKF).

Measurements with similar orientations were removed to enhance the quality of the datasets. A small angular threshold was used to compare the unit vectors to avoid redundant directions. The number of representative points per direction was only one. This minimized the overrepresentation of some orientations and avoided bias in the ellipsoid fitting.

The results confirm that ellipsoid calibration significantly improves measurement consistency. Although the calibrated data does not perfectly match theoretical gravity, the remaining error is consistent with the limitations of low-cost MEMS IMUs.

5.3. Gravity Compensation and Motion Analysis

A segment of real-world experimental data was collected during a USV deployment. Both acceleration and velocity measurements were recorded.

A low-pass filter with a cutoff frequency of 1 Hz was applied to the raw accelerometer data. Then the data were calibrated using the ellipsoid fitting procedure. Measurements with similar orientations were removed to reduce bias in the calibration process. The calibrated acceleration was then rotated to the NED frame using the recorded attitude. Gravity was subtracted to obtain the estimated acceleration, denoted as $\mathbf{a}_{\text{est}}$.

A reference acceleration, $\mathbf{a}_{\text{ref}}$, was computed from the recorded velocity by numerical differentiation. To reduce numerical differentiation noise, a second-order Butterworth low-pass filter with a cutoff frequency of 1 Hz was applied to the velocity measurements prior to differentiation. The forward Euler approximation was applied with the sample interval of the acquisition system:

$$\mathbf{a}_{\text{ref}}(t_i) = \frac{v(t_{i+1}) - v(t_{i-1})}{2\Delta t} \tag{31}$$

Both $\mathbf{a}_{\text{est}}$ and $\mathbf{a}_{\text{ref}}$ were compared along the x, y, and z axes. The trends of $\mathbf{a}_{\text{est}}$ closely followed $\mathbf{a}_{\text{ref}}$, indicating that the calibration and gravity compensation can effectively capture the vehicle motion. Figure 10 shows the comparison of acceleration along different axes between $\mathbf{a}_{\text{est}}$ and $\mathbf{a}_{\text{ref}}$, and Figure 11 illustrates the change in $\left\|\mathbf{a}_{\text{est}}\right\|$ and $\left\|\mathbf{a}_{\text{ref}}\right\|$. The blue line and red line represent $\mathbf{a}_{\text{est}}$ and $\mathbf{a}_{\text{ref}}$, respectively. Figure 12 shows the acceleration error over time.

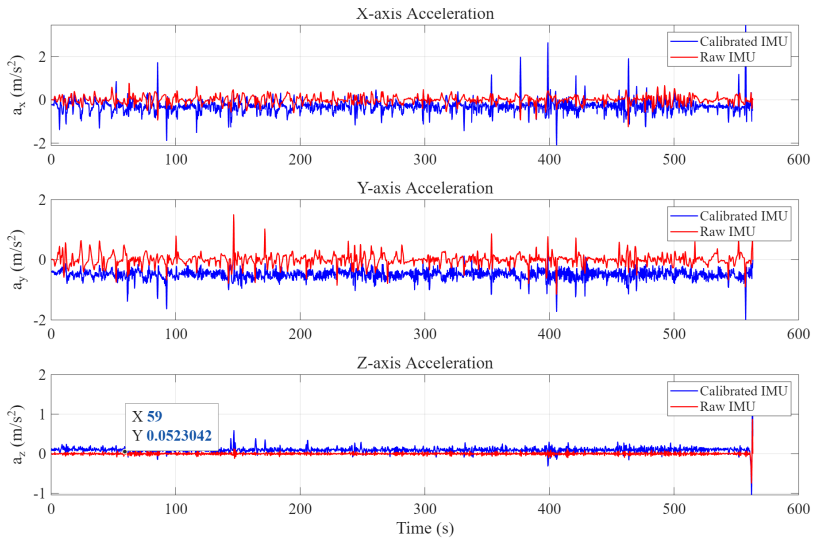


Figure 10. ax, ay and az vs. Time.

The mean absolute error between the estimated acceleration and the reference acceleration was calculated as 0.4423 m/s^2. Minor differences were observed, mainly related to sensor noise, small errors in attitude estimation, and numerical differentiation errors used to obtain reference acceleration.

The accuracy of gravity compensation strongly depends on the attitude estimation provided by the EKF3 module of ArduSub. To quantify this effect, a first-order error propagation analysis is performed.

In the Body Frame, the acceleration measured by the IMU, a_m, contains the vehicle's true linear acceleration a_{est} and the projection of the gravity vector in the body frame g_b:

$$a_m = a_{est} - g_b \tag{32}$$

Using Equation (1), g_b can be calculated as:

$$g_b = R^{-1}\begin{bmatrix}0\\0\\g\end{bmatrix} = \begin{bmatrix}-g\ sin\ \theta\\ g\ sin\ \phi\ cos\ \theta\\ g\ cos\ \phi\ cos\ \theta\end{bmatrix} \tag{33}$$

Therefore, the compensated estimated linear acceleration is:

$$a_{est} = \begin{bmatrix}a_{est,x}\\a_{est,y}\\a_{est,z}\end{bmatrix} = \begin{bmatrix}a_{m,x} + g\ sin\ \theta\\ a_{m,y} - g\ sin\ \phi\ cos\ \theta\\ a_{m,z} - g\ cos\ \phi\ cos\ \theta\end{bmatrix} \tag{34}$$

Taking the partial derivatives with respect to ϕ and θ yields the first-order Taylor expansion for error propagation:

$$\begin{bmatrix}\delta a_{est,x}\\ \delta a_{est,y}\\ \delta a_{est,z}\end{bmatrix} = \begin{bmatrix}0 & g\ cos\ \theta\\ -g\ cos\ \phi\ cos\ \theta & g\ sin\ \phi\ sin\ \theta\\ g\ sin\ \phi\ cos\ \theta & g\ cos\ \phi\ sin\ \theta\end{bmatrix}\begin{bmatrix}\delta\phi\\ \delta\theta\end{bmatrix} \tag{35}$$

USV operates in the water with a nearly horizontal attitude ($\phi \approx 0^\circ$, $\theta \approx 0^\circ$). Under these conditions, the sensitivity matrix is greatly simplified:

$$\delta a_{est,x} \approx g \cdot \delta\theta, \tag{36}$$

$$\delta a_{est,y} \approx -g \cdot \delta\phi, \tag{37}$$

$$\delta a_{est,z} \approx 0, \tag{38}$$

Under typical dynamic wave conditions, EKF3 attitude errors are within $\pm 1^\circ$ ($\approx$0.017 rad). Therefore, the corresponding acceleration uncertainty introduced in gravity compensation is approximately:

$$\delta a \approx g \cdot 0.017 \approx 0.167 \tag{39}$$

These observations suggest that, after calibration and gravity compensation, the accelerometer data can provide a practical approximation of the vehicle's linear acceleration. The results provided in the comparison with acceleration gained from velocity validate the usability of the pipeline in the short-term tracking of movements and estimation of accelerations in real-life USV work.

Recent USV navigation systems are mainly based on GNSS/INS integration and EKF-based state estimation frameworks, where IMU measurements are typically corrected using factory calibration or simple bias compensation, rather than full intrinsic calibration modeling. For instance, recent marine robotic and USV platforms rely on multi-sensor fusion to improve state estimation robustness [23–26]. As a result, residual acceleration errors under dynamic sea conditions remain in the range of approximately 0.1–0.8 m/s^2 in similar systems [27–29]. Hence, the proposed ellipsoid-based calibration significantly improves the consistency of acceleration measurements.

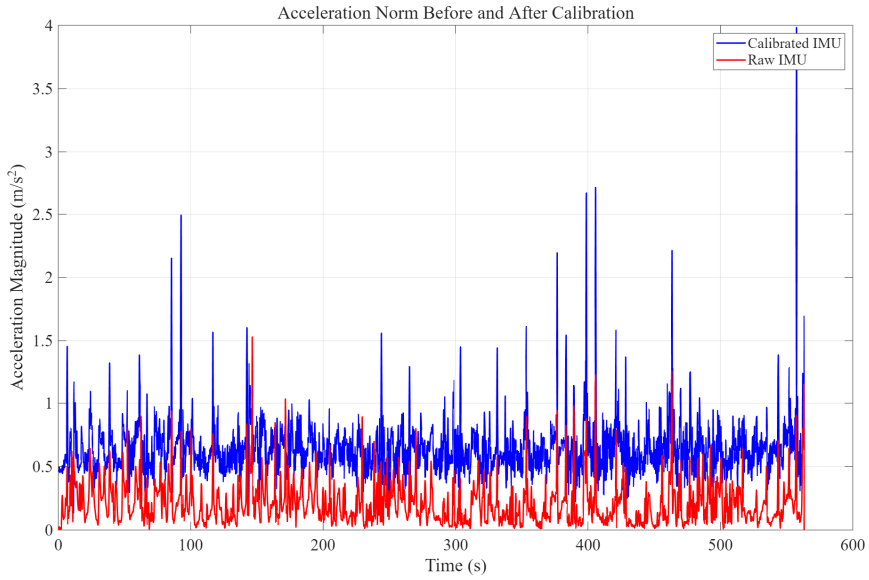


Figure 11. Acceleration vs. Time.

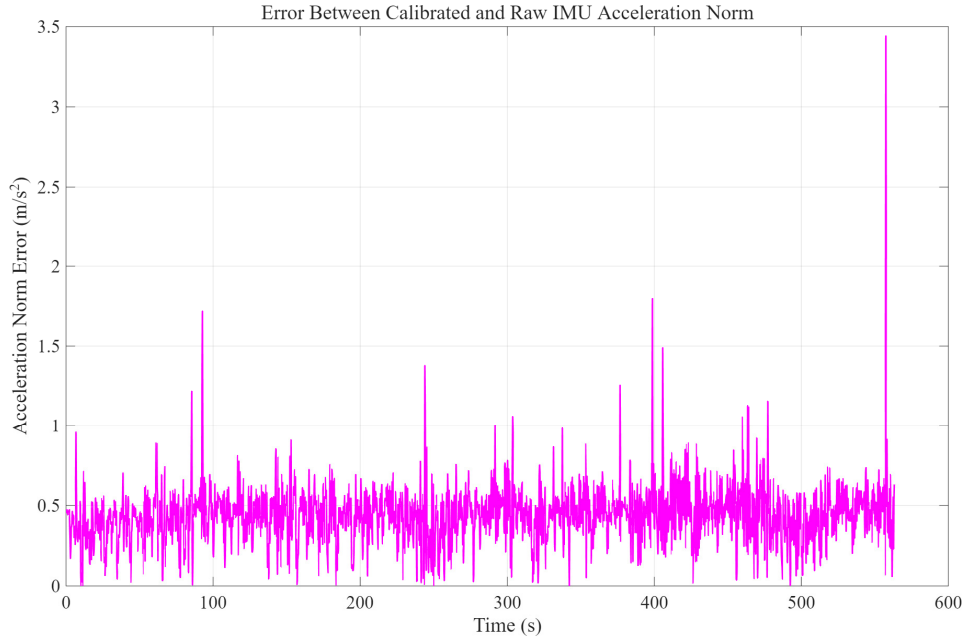


Figure 12. Acceleration Error vs. Time.

6. Discussion

This report concentrated on the design, control, and testing of a USV using the ArduSub firmware. The main aim was to have a stable system architecture that incorporates vehicle control, data transmission, and onboard sensing, and test its functionality in actual working conditions.

The experimental findings demonstrate that the USV platform built around ArduSub offers stable low-level control combined with dependable telemetry for surface operation. The integration of the Pixhawk flight controller, communication via MAVLink, and remote joystick input allowed for the execution of all commands in real time and sensor feedback. The control commands reflected in vehicle motion were appropriate in both the static and dynamic tests, which made it possible to carry out meaningful motion analysis.

IMU calibration and acceleration processing were also presented as a tool for vehicle motion assessment rather than a standalone sensor experiment. Accelerometer bias and

scale factor error were partially corrected through ellipsoid fitting in simulation and through static experiments. In the simulation test, the mean value of the acceleration excluding gravity was 0.1610 m/s^2 compared to 0.7972 m/s^2 before calibration. In the real static test, the average value of the magnitude of acceleration when gravity was removed was reduced from 0.29 m/s^2 to 0.26 m/s^2. This indicates the inherent limitations of low-cost MEMS sensors deployed on embedded marine systems.

In real-world implementation, the measurements of the acceleration were calibrated and converted to the NED frame with gravity removed using attitude estimates of the ArduSub EKF3. The acceleration estimates of the resulting linear acceleration were compared with the reference acceleration that was derived from velocity measurements. The mean absolute error between the estimated and reference accelerations was found to be 0.4423 m/s^2. The estimated acceleration closely matched the general direction of the reference data, which indicates that the sensing and processing pipeline captured the main characteristics of the vehicle motion. The main causes of minor differences were sensor noise, slight IMU misalignment, error in attitude estimation, and numerical differentiation of velocity.

Data transmission was also crucial during system validation. Acceleration, velocity, and attitude data can all be gathered in real time due to reliable underwater telemetry communication. This enabled the direct comparison of commanded motion, measured velocity, and estimated acceleration to show that the USV system is a coherent sensing–control–communication system.

While the current system demonstrates stable operation and reliable sensing performance, several directions remain for further improvement and scalability. First, the calibration procedure can be extended to include online or adaptive calibration, allowing for compensation for time-varying bias and environmental effects during long-duration missions. Second, the accuracy of gravity compensation can be further improved by incorporating higher-precision attitude estimation, such as multi-sensor fusion to reduce error propagation from orientation uncertainty. Third, the current acceleration validation relies on velocity differentiation; future work will explore independent reference systems such as a Doppler Velocity Log or external tracking systems to provide more accurate ground truth. In addition, robust filtering and smoothing techniques could be introduced to mitigate noise amplification in dynamic conditions. From a system perspective, challenges remain in scaling the platform to long-duration deployments, including power management, communication reliability in harsh marine environments, and robustness under stronger wave disturbances. Addressing these aspects will be essential for deploying the system in real-world autonomous navigation and marine monitoring applications.

In general, the findings indicate that the proposed ArduSub-based USV platform can operate reliably, transmit data reliably, and enable meaningful motion estimation. This proposed USV is applicable to future navigation and control efforts as well.

7. Conclusions

An ArduSub-based surface vehicle was successfully implemented and experimentally tested. The system combines low-level vehicle control and a real-time data transmission system with onboard inertial sensing in a single architecture. Both simulation and experimental studies were conducted to evaluate the performance of the system and the reliability of the data.

The results show that the USV platform achieved stable surface operation and reliable telemetry transmission. IMU calibration and gravity compensation enhanced the stability of the acceleration measurements and enabled reliable estimation of linear acceleration in the actual deployment. Moreover, the acceleration derived from velocity showed good

agreement with the processed inertial measurements, indicating that the inertial data reliably represent the actual vehicle motion.

Despite the residual errors caused by the limitations of low-cost sensors and the accuracy of attitude estimators, the overall system performance is adequate for performing short-term motion analysis, estimating velocity, and validating system performance experimentally. The results demonstrate that ArduSub can be easily adapted to fit the USV applications and offers a versatile platform for research and development.

Future work is aimed at the extension of offshore experiments, enhancement of filtering and sensor fusion, and incorporation of navigation and control algorithms. These improvements are expected to enhance the motion estimation precision and expand the functionality of the USV platform.

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Abbreviations

The following abbreviations are used in this manuscript:

USV	Unmanned Surface Vehicle
GNSS	Global Navigation Satellite System
MEMS	Micro-Electro-Mechanical Systems
NED	North–East–Down
CG	center of gravity
EVD	eigenvalue decomposition

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